

CSV 1 - airline_losses

CSV 1 — `airline_losses.csv`

Step	Validation / Cleaning Activity	Result	Action Taken
1	Imported dataset using <code>pandas.read_csv()</code>	Successful	Dataset loaded into DataFrame
2	Verified dataset dimensions	70 rows × 8 columns	No action required
3	Reviewed column names	All columns correctly identified	No action required
4	Validated data types (<code>df.info()</code>)	4 Text (<code>str</code>), 2 Integer (<code>int64</code>), 2 Decimal (<code>float64</code>)	No datatype conversion required
5	Checked duplicate records	0 duplicate rows	No records removed
6	Checked missing values	0 missing values across all columns	No imputation required
7	Validated text columns	<code>airline</code> , <code>country</code> , <code>airline_type</code> , <code>region</code> identified correctly	No action required
8	Checked leading/trailing whitespace	No whitespace detected	No trimming required
9	Reviewed unique category values	Categories were consistent with no spelling or formatting issues	No standardization required
10	Reviewed category distribution	Distribution consistent with dataset design	No action required
11	Generated descriptive statistics (<code>describe()</code>)	Numeric ranges and summary statistics appeared valid	No correction required
12	Exported validated dataset	Clean dataset created	Saved as <code>airline_losses_clean.csv</code>

Final Outcome

Dataset Status: Passed all validation checks.

Cleaning Performed: None required.

Reason: Source dataset was already analytically clean.

CSV 2 - airline_losses_estimate

CSV 2 — `airline_losses_estimate.csv`

Step	Validation / Cleaning Activity	Result	Action Taken
1	Imported dataset using <code>pandas.read_csv()</code>	Successful	Dataset loaded into DataFrame
2	Verified dataset dimensions	35 rows × 7 columns	No action required
3	Reviewed column names	All columns correctly identified	No action required
4	Validated data types (<code>df.info()</code>)	2 Text (<code>str</code>), 5 Integer (<code>int64</code>)	No datatype conversion required
5	Checked duplicate records	0 duplicate rows	No records removed
6	Checked missing values	0 missing values across all columns	No imputation required
7	Validated text columns	<code>airline</code> , <code>country</code> identified correctly	No action required
8	Checked leading/trailing whitespace	No whitespace detected	No trimming required
9	Reviewed unique category values	Categories were consistent; no spelling or formatting inconsistencies detected	No standardization required
10	Reviewed category distribution	Distribution consistent with source dataset	No action required
11	Reviewed country values	<code>Scandinavia</code> retained exactly as provided in the source dataset	No modification made to avoid altering source data
12	Generated descriptive statistics (<code>describe()</code>)	Numeric ranges and summary statistics appeared valid	No correction required
13	Exported validated dataset	Clean dataset created	Saved as <code>airline_losses_estimate_clean.csv</code>

Final Outcome

Dataset Status: Passed all validation checks.

Cleaning Performed: None required.

Reason: Source dataset required validation only; no transformations were necessary.

CSV 3 – airport_disruptions

CSV 3 – airport_disruptions.csv.

Step	Validation / Cleaning Activity	Result	Action Taken
1	Imported dataset using <code>pandas.read_csv()</code>	Successful	Dataset loaded into DataFrame
2	Verified dataset dimensions	227 rows × 9 columns	No action required
3	Reviewed column names	All columns correctly identified	No action required
4	Validated data types (<code>df.info()</code>)	6 Text (<code>str</code>), 1 Integer (<code>int64</code>), 1 Float (<code>float64</code>), 1 Date stored as <code>str</code>	Converted <code>date</code> column to <code>datetime64[ns]</code> using <code>pd.to_datetime()</code>
5	Checked duplicate records	0 duplicate rows	No records removed
6	Checked missing values	0 missing values across all columns	No imputation required
7	Validated text columns	<code>airport_name</code> , <code>iata_code</code> , <code>country</code> , <code>region</code> , <code>disruption_type</code> , <code>severity_level</code> identified correctly	No action required
8	Checked leading/trailing whitespace	No whitespace detected	No trimming required
9	Reviewed unique category values	Categories were consistent; no spelling or formatting inconsistencies detected. Sao and	No standardization required
10	Reviewed category distribution	Distribution consistent with source dataset	No action required

11	Validated date values	Date values successfully converted to datetime format	Datetime conversion completed for analytical compatibility
12	Generated descriptive statistics (<code>describe()</code>)	Numeric ranges and summary statistics appeared valid	No correction required
13	Exported validated dataset	Clean dataset created	Saved as <code>airport_disruptions_clean.csv</code>

Final Outcome

Dataset Status: Passed all validation checks.

Cleaning Performed: Converted the `date` column from string (`str`) to `datetime64[ns]`.

Reason: Datetime conversion enables accurate date filtering, SQL date functions, time-series analysis, and Power BI date intelligence. Everything else in the source dataset was already clean and required validation only.

CSV 4 - airspace_closures

CSV 4 — airspace_closures.csv

Step	Validation / Cleaning Activity	Result	Action Taken
1	Imported dataset using <code>pandas.read_csv()</code>	Successful	Dataset loaded into DataFrame
2	Verified dataset dimensions	84 rows × 8 columns	No action required
3	Reviewed column names	All columns correctly identified	No action required
4	Validated data types (<code>df.info()</code>)	5 Text (<code>str</code>), 1 Integer (<code>int64</code>), 2 Date-Time columns stored as text (<code>str</code>)	Converted <code>closure_start_date</code> and <code>closure_end_date</code> to <code>datetime64[ns]</code> using <code>pd.to_datetime()</code>
5	Checked duplicate records	0 duplicate rows	No records removed
6	Checked missing values	0 missing values across all columns	No imputation required
7	Validated text columns	<code>country</code> , <code>region</code> , <code>airspace_zone</code> , and <code>reason</code> identified correctly	No action required
8	Checked leading/trailing whitespace	No whitespace detected	No trimming required
9	Reviewed unique category values	Categories were consistent; no spelling or formatting inconsistencies detected	No standardization required
10	Reviewed category distribution	Distribution consistent with source dataset	No action required

11	Generated descriptive statistics (<code>describe()</code>)	Numeric ranges and summary statistics appeared valid	No correction required
12	Exported validated dataset	Clean dataset created	Saved as <code>airspace_closures_clean.csv</code>

Final Outcome

Dataset Status: Passed all validation checks.

Cleaning Performed: Converted `closure_start_date` and `closure_end_date` from text (`str`) to `datetime64[ns]`.

Reason: Preserved complete timestamp information for PostgreSQL and Power BI while enabling SQL date/time functions, filtering, sorting, grouping, and time-series analysis without creating redundant date and time columns.

CSV 5 — conflict_events

CSV 5 — conflict_events.csv

Step	Validation / Cleaning Activity	Result	Action Taken
1	Imported dataset using <code>pandas.read_csv()</code>	Successful	Dataset loaded into DataFrame
2	Verified dataset dimensions	82 rows × 6 columns	No action required
3	Reviewed column names	All columns correctly identified	No action required
4	Validated data types (<code>df.info()</code>)	6 Text (<code>str</code>) columns	Converted <code>date</code> column from <code>str</code> to <code>datetime64[ns]</code> using <code>pd.to_datetime()</code>
5	Checked duplicate records	0 duplicate rows	No records removed
6	Checked missing values	0 missing values in the original dataset	No imputation required
7	Validated text columns	All text columns identified correctly	No action required
8	Checked leading/trailing whitespace	No whitespace detected	No trimming required
9	Reviewed unique category values	Categories were consistent; no spelling or formatting inconsistencies detected	No standardization required
10	Reviewed category distribution	Distribution consistent with source dataset	No action required
11	Normalized <code>location</code> column	Composite values identified (e.g., <code>City, Country</code>)	Split into <code>location_name</code> and <code>country</code> columns using <code>str.split()</code>

12	Trimmed whitespace after split	No leading/trailing whitespace detected	Applied <code>.str.strip()</code> to <code>location_name</code> and <code>country</code> for consistency
13	Removed redundant <code>location</code> column	Normalized schema achieved	Dropped original <code>location</code> column after successful split
14	Verified transformed dataset	Data types and schema confirmed	Reviewed updated <code>df.info()</code> after transformations
15	Generated descriptive statistics (<code>describe()</code>)	Date summary statistics generated successfully	No correction required
16	Exported validated dataset	Clean dataset created	Saved as <code>conflict_events_clean.csv</code>

Final Outcome

Dataset Status: Passed all validation checks.

Cleaning Performed:

- Converted `date` from text (`str`) to `datetime64[ns]`.
- Split the composite `location` field into `location_name` and `country`.
- Trimmed whitespace from the newly created columns.
- Removed the original `location` column after successful normalization.

Reason: The dataset was normalized to improve SQL querying, filtering, grouping, and Power BI reporting while preserving the original location information in a structured format.

CSV 6 — flight_cancellations

CSV 6 — flight_cancellations.csv

Step	Validation / Cleaning Activity	Result	Action Taken
1	Imported dataset using <code>pandas.read_csv()</code>	Successful	Dataset loaded into DataFrame
2	Verified dataset dimensions	2200 rows × 12 columns	No action required
3	Reviewed column names	All columns correctly identified	No action required
4	Validated data types (<code>df.info()</code>)	<code>date</code> stored as text (<code>str</code>); remaining columns already had appropriate data types	Converted <code>date</code> to <code>datetime64[ns]</code>
5	Checked duplicate records	0 duplicate rows	No records removed
6	Checked missing values	0 missing values	No imputation required
7	Validated text columns	All text columns identified correctly	No action required
8	Checked leading/trailing whitespace	No whitespace detected	No trimming required
9	Reviewed unique category values	Categories consistent; no spelling or formatting inconsistencies detected	No standardization required
10	Reviewed category distribution	Distribution consistent with source dataset	No action required
11	Generated descriptive statistics (<code>describe()</code>)	Numeric ranges and summary statistics appeared valid	No correction required

12	Exported validated dataset	Clean dataset created	Saved as flight_cancellations_clean.csv
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Final Outcome

Dataset Status: Passed all validation checks.

Cleaning Performed:

- Converted `date` from text (`str`) to `datetime64[ns]`.

Reason: Source dataset required validation only; no additional transformations were necessary.

CSV 7 — flight_reroutes

CSV 7 — flight_reroutes.csv

Step	Validation / Cleaning Activity	Result	Action Taken
1	Imported dataset using <code>pandas.read_csv()</code>	Successful	Dataset loaded into DataFrame
2	Verified dataset dimensions	1500 rows × 16 columns	No action required
3	Reviewed column names	All columns correctly identified	No action required
4	Validated data types (<code>df.info()</code>)	<code>date</code> stored as text (<code>str</code>); remaining numeric columns already had appropriate data types	Converted <code>date</code> to <code>datetime64[ns]</code>
5	Checked duplicate records	0 duplicate rows	No records removed
6	Checked missing values	0 missing values	No imputation required
7	Validated text columns	All text columns identified correctly	No action required
8	Checked leading/trailing whitespace	No whitespace detected	No trimming required
9	Reviewed unique category values	Categories consistent; no spelling or formatting inconsistencies detected	No standardization required
10	Reviewed category distribution	Distribution consistent with source dataset	No action required
11	Generated descriptive statistics (<code>describe()</code>)	Numeric ranges and summary statistics appeared valid	No correction required
12	Exported validated dataset	Clean dataset created	Saved as <code>flight_reroutes_clean.csv</code>

Final Outcome

Dataset Status: Passed all validation checks.

Cleaning Performed:

- Converted `date` from text (`str`) to `datetime64[ns]`.

Reason: Source dataset required validation only; all remaining columns were already stored using appropriate data types.